MILITARY SPECIFICATION SHEET

ATTENUATORS, VARIABLE (COAXIAL, STEP)
CLASS 5 (SMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the attenuators described herein shall consist of this document and the latest issue of Specification MIL-A-24215

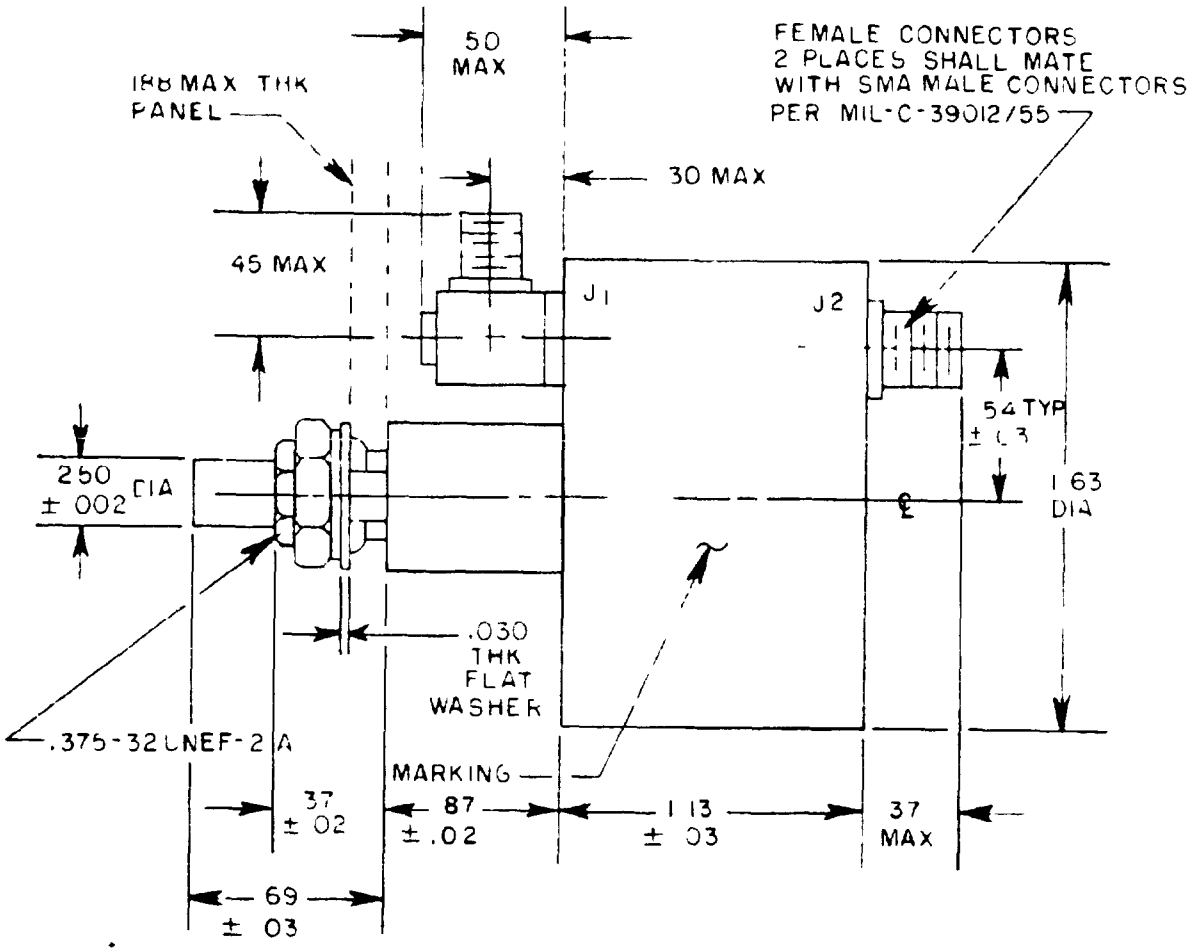
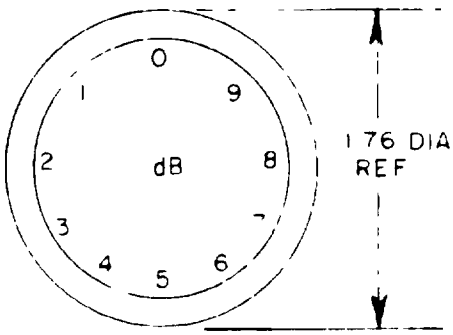


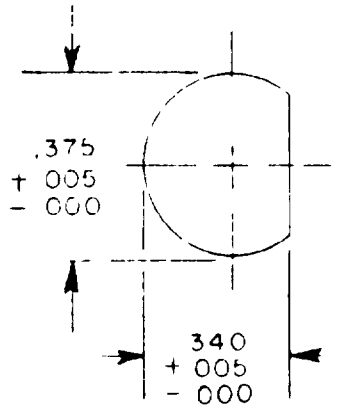
FIGURE 1 Dimensions and configuration for part number M24215/13-01

FSC 5985

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SHAFT KNOB



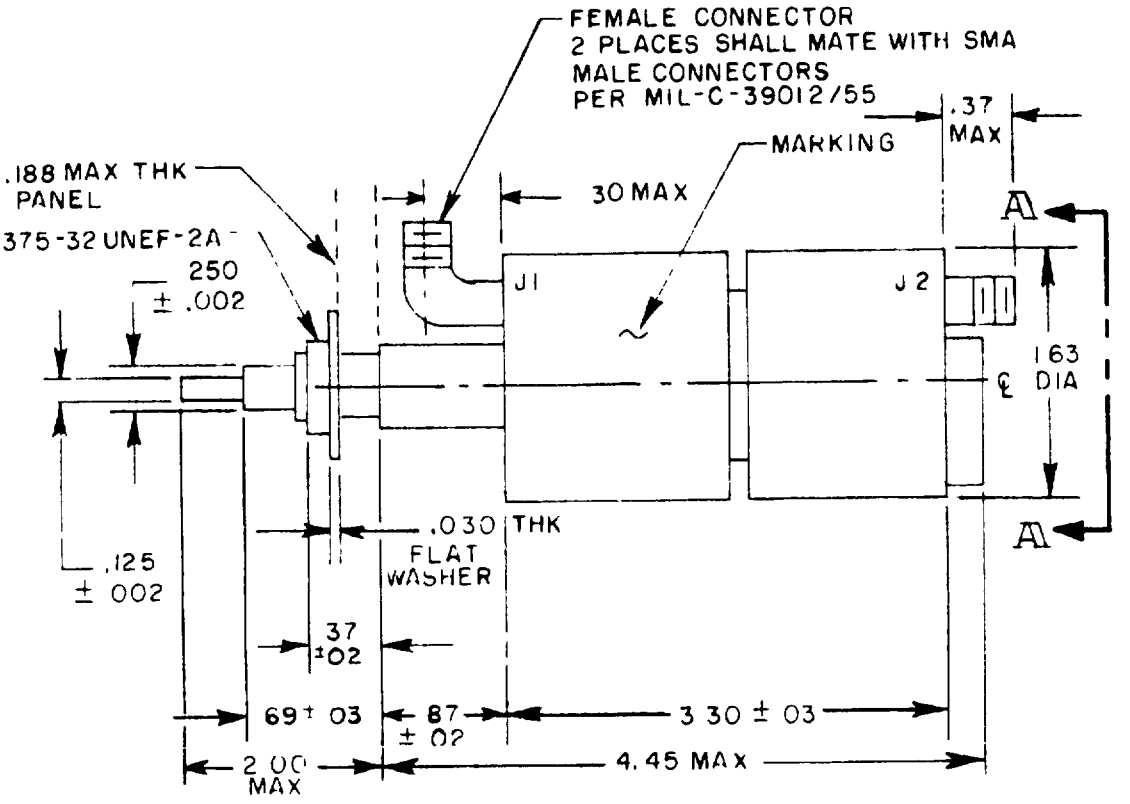
PANEL MOUNTING HOLE

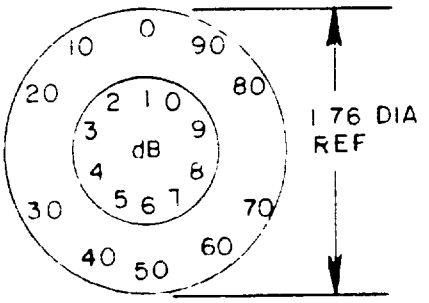
INCHES	MM
005	13
02	5
03	8
188	4 78
.250	6.35
30	7 6
340	8 64
37	9 4
375	9.52
45	11.4
50	12 7
54	13 7
69	17 5
.87	22 1
1 13	28 7
1.63	41.4
1 76	44 7

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

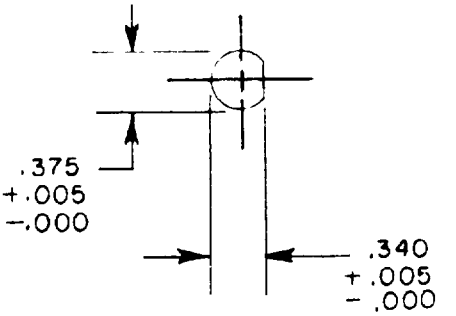
FIGURE 1 Dimensions and configuration for part number M24215/13-01 - Continued

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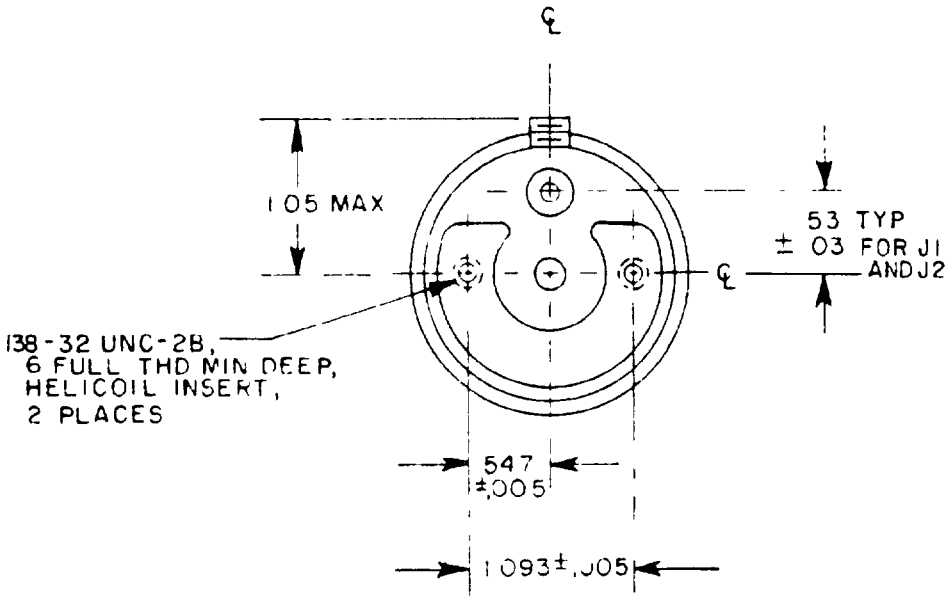
SHAFT KNOBS



PANEL MOUNTING HOLE

FIGURE 2 Dimensions and configuration for part number M24215/13-02

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VIEW A-A

INCHES	MM
002	05
.005	13
.02	5
030	76
125	3 18
188	4 78
250	6 35
30	7 6
340	8 64
37	9 4
375	9 52
53	13 5
547	13 89
69	17 5
87	22 1
1 05	26 7
1 093	27 76
1 63	41 4
1 76	44 7
2 00	50 8
3 30	83 8
4 45	113 0

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 2 Dimensions and configuration for part number M24215/13-02 - Continued

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TABLE I Electrical requirements

Part number M24215/13	Frequency range (GHz)	Attenuation range (dB)	Accuracy	VSWR (max)	Insertion loss (max)	Weight (max) oz
01	DC-18	0-9 in 1 dB steps	For 1-9 dB DC- 1.0 ±0.2 dB DC-12 4 ±0 3 dB DC-18 0 ±0 5 dB	1 5 1 2/	DC- 1 0 GHz 0 2 dB DC- 4 0 GHz 0 3 dB DC- 8 0 GHz 0 5 dB DC-12 4 GHz 0 7 dB DC-18.0 GHz 1.0 dB	3.5
02	DC-4	1/ 0-99 in 1 dB steps	1- 9 dB ±0 3 dB 10-19 dB ±0 7 dB 20-29 dB ±0 9 dB 30-39 dB ±1 0 dB 40-49 dB ±1 1 dB 50-59 dB ±1 2 dB 60-69 dB ±1 3 dB 70-79 dB ±1 4 dB 80-89 dB +1 5 dB 90-99 dB ±1.6 dB	1 35 1	DC-4 GHz 0.6 dB	11

1/ Two controls, 0 to 9 in 1 dB steps and 0 to 90 in 10 dB steps, rotate independently, with stops between 0 and 90 dB

2/ Each attenuator shall be supplied with plots for insertion loss and VSWR The insertion loss plot shall not deviate more than ±0 10 dB from a straight line between 1.0 MHz and 18 0 GHz The VSWR plot shall be a family of curves depicting the VSWR for each attenuator step These tests shall be conducted at 70°C

REQUIREMENTS

Dimensions and configuration See figures 1 and 2.

Input and output impedance 50 ±2 ohms

Frequency sensitivity ±0.05 dB

Electrical requirements See table I These requirements apply with either RF connector used as the input

Weight See table I

Input power (minimum) 2 watts average, 200 watts peak, at any setting.

Temperature range
Operating 0° to 70°C
Nonoperating -62° to +85°C.

Temperature coefficient 0001 dB/dB/°C

Material
Housing and drum. Aluminum alloy
Control shaft Stainless steel.

Finish
Aluminum parts Shall be protected with a chemical film treatment in accordance with MIL-C-5541, class 3
Stainless steel parts Shall be passivated in accordance with MIL-S-5002

Environmental All the environmental requirements of MIL-A-24215 apply, except
Shock Shall be in accordance with MIL-STD-202, method 213, condition J, except 8 G, 100 msec
Vibration Shall be in accordance with MIL-STD-202, method 204, except the level and frequencies of figure 3 shall be used

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Burn-in Each dash number 01 delivered on contract shall be subjected to a burn-in of 96, -0 +8 hours under the following conditions

A RF input power 1 watt average, minimum.
B. Temperature 70° ±1°C
C Input frequency at start and through 96 hours shall be as follows
1. 2 GHz, ±500 MHz, 0 to 11, ±1 hour
2 4 GHz, ±500 MHz, 11 to 22, ±1 hour.
3 6 GHz, ±500 MHz, 22 to 33, ±1 hour
4 8 GHz, ±500 MHz, 33 to 44, ±1 hour.
5. 10 GHz, ±500 MHz, 44 to 55, ±1 hour
6 12 GHz, ±500 MHz, 55 to 66, ±1 hour.
7 14 GHz, ±500 MHz, 66 to 77, ±1 hour
8 16 GHz, ±500 MHz, 77 to 88, ±1 hour.
9 18 GHz, ±500 MHz, 88 to 96, ±1 hour

During this time period the attenuator shall be cycled, starting with 0 dB through the maximum attenuation, then repeated for a total of 1152 steps in even increments, over the 96 hour burn-in period At the completion of this burn-in, attenuators shall meet the electrical requirements of table I

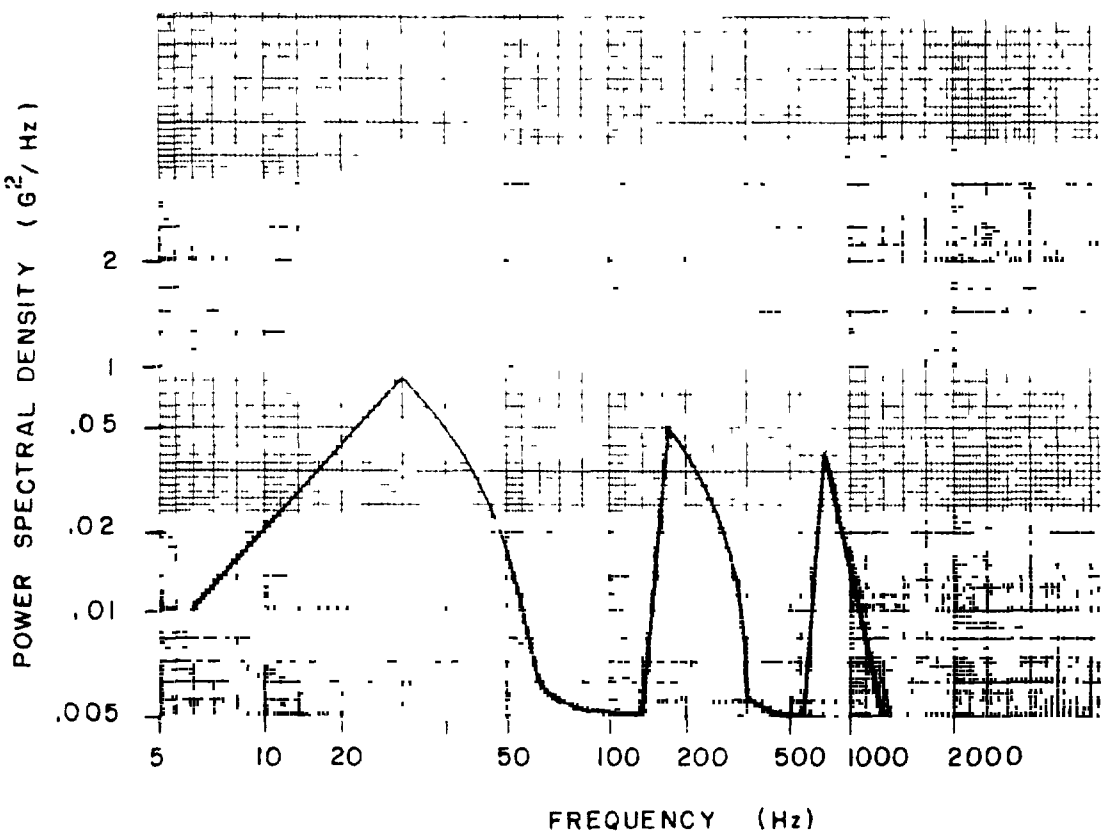


FIGURE 3 Vibration level

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Part number See table 1 and figure 1 and 2

NOTE Attenuators covered by this specification sheet are substituted for the following manufacturer's part number. This information is no way implies that the manufacturer's part is suitable as a substitute for the military part number

Dash number	Manufacturer's code	Manufacturer's part number	FSN
01	12436	6010 202-001	
01	93459	AF-9009	
02	05869	721733	
02	93459	9063A	

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR, MI, SL
Air Force - 11
DLA - ES

User activities
Army - SM
Navy - AS, MC, OS, SH

Preparing activity
Navy - EC

Agent.
DLA - ES

(Project 5985-0928-2)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

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DOCUMENT IDENTIFIER AND TITLE
MIL-A-24215/13 ATTENUATORS, VARIABLE (COAXIAL, STEP) CLASS 5 (SMAO

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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